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MOOCS Decision Modeling Course

Waiting Lines (Queuing Theory)

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Queuing Theory - Introduction

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Queuing Theory

- **Waiting Lines or Queues** are everywhere. Examples could be found in banks. Railway ticket counters, movie ticket counters, post offices and so on.
- Waiting in queues leads to **tremendous loss of time** – Americans lose nearly 20 million person-years per year waiting in queues.
- **Also consider**:
  - Aeroplanes waiting for runway
  - Machines waiting for repair
  - Vehicles waiting in traffic jam
  - Ships waiting in sea for entry into the port
- **Queuing theory** is the study of waiting in many different situations.
- **Queuing models** determine how to operate a queuing system effectively balancing cost of waiting and that of providing service.

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Applications of Queuing Theory

Commercial

- ATM/ Bank/ Repair Shop
- Vending machine/ Payment for purchase

Transportation

- Cars in traffic / Ship unloading
- Elevators / parking Lot

Business – Industrial

- Material handling / Maintenance
- Health services/ Job shop
- Telecommunications

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Basic Queuing Process

Basic Queuing Process

Input Source: it generates the customers over time

Queuing System: Customers so generated enter the queuing system

Queue: The queuing systems contains a queue for customers to wait

Queue Discipline: A member of the queue is selected for service by some rule known as the queue discipline.

Service Mechanism: The required service is provided by the service mechanism after which the customer leaves the queuing system.

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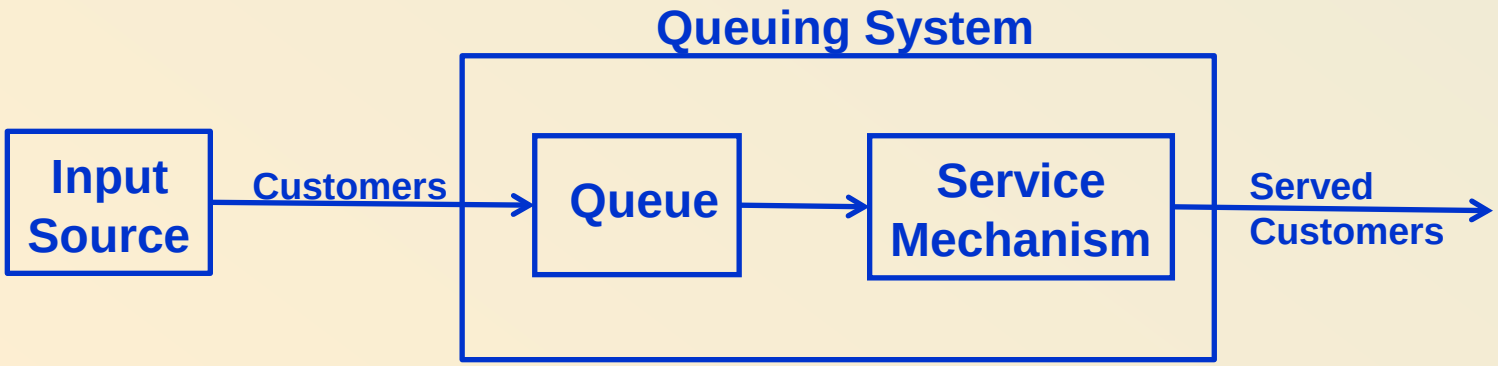

Input Source

- Input Source (Calling Population)
- Total number of customers requiring service from time to time – also known as calling population.
- Size of input source is often assumed to be infinite – unlimited input source in order to simplify calculations.
- Number of customers generated until a specific time is usually assumed to follow a Poisson distribution – arrivals are random but at a certain fixed mean rate.
- An equivalent assumption is that the interarrival time is having an exponential distribution.

- Some special cases are as follows:
- *balking* where customers refuse to join a queue seeing its length
- *jockeying* where customers jumps from one queue to another.
- *reneging* where customers leave after spending some time in a queue before getting the service.

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Queuing Theory

- **Queue**
- **characterized by maximum number of customers that it can contain. It could be finite or infinite. Infinite assumption is more common.**

- **Queue Discipline**
- **Order in which customers are selected for service from the queue. Example: first-come-first-served (FCFS), random, Service in Random Order (SIRO), Last in first out (LIFO), or according to some priority procedure. FCFS is the default.**

- **Service Mechanism**
- **It consists of one or more parallel service channels known as *servers. service time of a customer is usually assumed to be exponentially distributed.* Other important service-time distributions are the *degenerate* distribution (constant service time) and the *Erlang* (gamma) distribution.**
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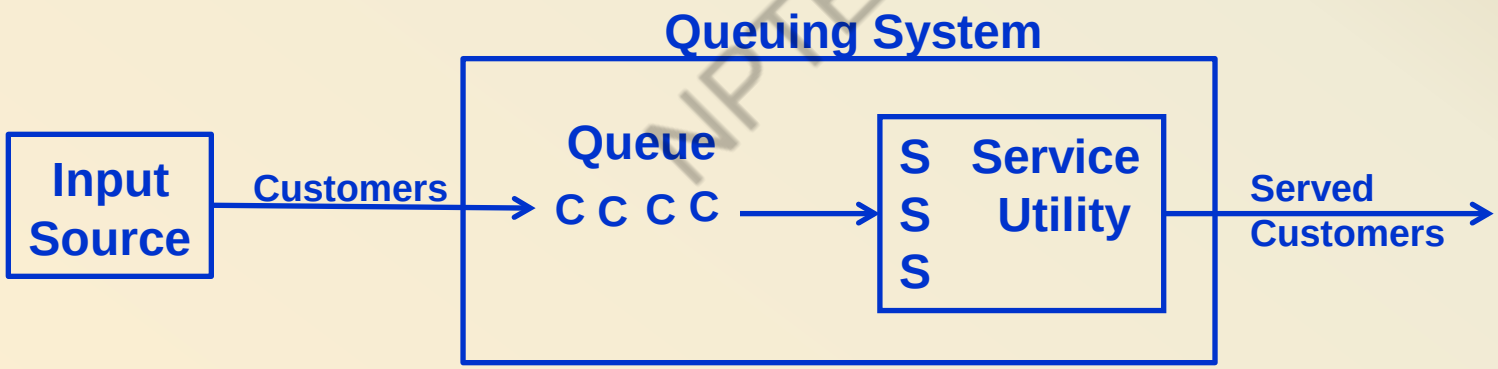
Elementary Queuing Process

An Elementary Queuing Process

- **A Single Waiting Line – empty at times**
- **A single service utility with one or more servers**
- **Each customer is served based on First Come First Serve**



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A Key Requirement

A key requirement for a queuing process is that

Arrival rate < Service rate

All the queuing relations are derived in the steady state.

Steady state will not be achieved if service rate is not higher than arrival rate!

For multiple servers, the combined service rare should be considered.

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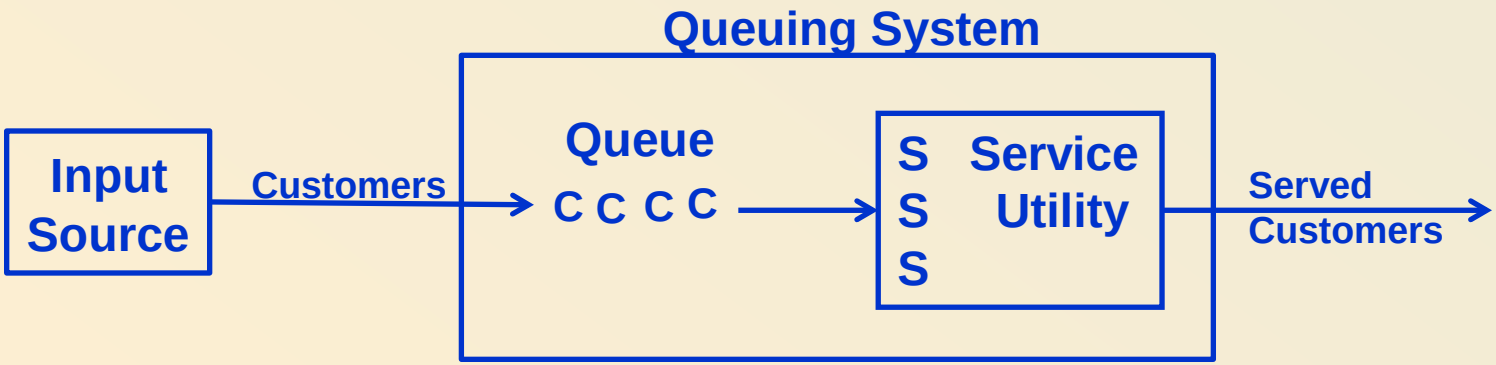

Steady State Performance Measures

- P_0 = Probability that there are no customers in the system
- P_n = Probability that there are "n" customers in the system
- L = Average number of customers in the system.
- L_q = Average number of customers in the queue.
- W = Average time a customer spends in the system.
- W_q = Average time a customer spends in the queue.
- ρ = Utilization rate for each server

  (the percentage of time that each server is busy).

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Little's Formula

- Little's Formulas represent important relationships between L, L_q, W, and W_q.

- These formulas apply to systems that meet the following conditions:
  - Single queue systems
  - Customers arrive at a finite arrival rate λ, and
  - The system operates under a steady state condition.

    - $L = \lambda W \qquad L_q = \lambda W_q$

    - Also, $W = W_q + 1/\mu$

- The formulas are quite robust and are applicable in many different queuing situations.

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Kendall Notation (a/b/c):(d/e/f)

To summarize the characteristics of the queuing system

Six parameters in shorthand: First three typically used, unless specified

- **a: Arrival Distribution (Poisson arrival rate)**
- **b: Service Distribution (Poisson departure/service rate)**
- **c: Number of servers (=1, 2,…, ∞)**
- **d: Service Discipline (FCFS/LIFO/SIRO, etc)**
- **e: Maximum number allowed in the system (infinite or finite)**
- **f: Size of input source (finite/infinite)**

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Classification of a Queuing Process

- Queuing system Classification example:
- M/M/1/FCFS/infinity/infinity e classified by:
  - M = Arrival process is Markovian
  - M = Service process is Markovian
  - 1 = Number of servers.
  - FCFS = Queue Discipline First Come First Serve
  - Infinity = System size (infinite waiting line).
  - Infinity = Population size.
- Notation
  - M (Markovian) = Poisson arrivals or exponential service time.
  - D (Deterministic) = Constant arrival rate or service time.
  - G (General) = General probability for arrivals or service time.

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Poisson and Exponential Distribution

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The Poisson Distribution

- Any random phenomenon that occurs on a per unit basis (per unit area, per unit volume, per unit time, etc.) is often well approximated by the Poisson distribution.

- Example: number of defective rivets in an airplane wing, spots per square meter of clothe, m/c break down per week, the number of misprints in a book, the number of customers arriving per day, Wrong phone calls received during a week

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The Arrival Process

- Arrivals could be deterministic or random.
- When arrivals are random, a usual assumption of the arrivals is the Poisson process.
- A Poisson Process has the following properties:
  - *Orderliness*: at most one customer will arrive during a given time interval.
  - *Stationarity*: probability of arrivals within a time interval is the same for all time intervals of equal length.
  - *Independence*: arrival of one customer has no influence on the arrival of another.

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Poisson Arrival Process

$$P(x=n) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$$

λ = mean arrival rate per time unit.

t = the length of the interval.

Customers arrive at a bank deposit counter according to a Poisson distribution at a rate of 6 per hour.

What is the probability that n customers will arrive in first half an hour (n = 0, 1, 2,…)?

Answer:

λ = 6; t = 1/2, hence, λt = 3.

P(n=0) = e^{-3} = 0.0498

P(n=1) = $3e^{-3}$ = 0.149

P(n=2) = $4.5\, e^{-3}$ = 0.224

P(n=3) = $4.5\, e^{-3}$ = 0.224

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Exponential Service Process

- Like arrivals, service also could be deterministic or random.
- When service process is random, a usual assumption of the service time is the exponential distribution.

$$f(t) = \mu e^{-\mu t}$$

- μ = average no of customers to be served per time period
- $1/\mu$ = the mean service time
- Probability that the service time X is less than time t;

$$P(x \leq t) = 1 - e^{-\mu t}$$

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Exponential Service Process

- **Probability that the service time X is less than time t;**

$$P(x \leq t) = 1 - e^{-\mu t}$$

If the average service time is $1/\mu = 6$ minutes per customer and Service time follows an exponential distribution.

What is the probability that it will take less than 3 minutes to serve the next customer?

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Answer:

The mean number of customers served per minute is 1/6 = 1/6(60) = 10 customers per hour.

- $P(x < .05 \text{ hours}) = 1 - e^{-(10)(.05)} = 1 - e^{-1/2} = 0.3935$

Six key Properties of Exponential Distribution

1) Strictly Decreasing function:

$$P(0 <= X <= \Delta t) > P(t <= X <= t + \Delta t)$$

2) Lack of Memory $P(X > t + \Delta t / X > \Delta t) = P(X > t)$

$$LHS = e^{-\lambda(t+\Delta t)} / e^{-\lambda(\Delta t)} = e^{-\lambda t} = RHS$$

3) Minimum of several independent exponential random variables has an exponential distribution

4) Relationship with Poisson distribution.

If No of arrivals is a Poisson process, then the distribution to inter-arrival time is exponential.

5) For all positive values of t, $P(X <= t + \Delta t / X > t) = \lambda \Delta t$ for small t

6) The distribution is unaffected by aggregation or disaggregation of the input process.

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Memory-less Property

Lack of Memory $P(X > t + \Delta t \mid X > \Delta t) = P(X > t)$

$$LHS = e^{-\lambda(t+\Delta t)} / e^{-\lambda(\Delta t)} = e^{-\lambda t} = RHS$$

For **exponential distribution**, probability of waiting a given length of time is **not affected** by how long one has already waited!

If someone has **already waited Δt minutes**, then probability that the person waits **another t minutes** is the same as that of someone else who is going to **wait t minutes** from the beginning.

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Suppose in a parking lot on a Sunday, last arrival was at 12 noon. λ = 1/hour. Inter-arrival time is exponential. **Nobody has arrived till 1 PM.** What is probability that nobody arrives before 2 PM?

Method 1: $P(X > 2 \mid X > 1) = e^{-1*2} / e^{-1*1} = e^{-2} / e^{-1} = e^{-1} = 0.368$

Method 2: As nobody arrived till 1 PM, Using **memory-less** property of exponential distribution: $e^{-1} = 0.368$

Queuing Example

In a parking lot on a Sunday, last arrival was at 12 noon. Arrival rate λ = 1/hour; service rate μ = 2/hour. No. of arrivals is Poisson and Inter-arrival time is exponential. Service time is also exponential.

a) What is probability that next arrival is before 1 PM?

$\lambda = 1$, $t = 1$, So, $\lambda t = 1$. reqd. prob. = $P(x \le t) = 1 - e^{-\lambda t} = 1 - e^{-1} = 0.632$

b) What is probability that next arrival is between 1 and 2 PM?

= Prob of arrival before 2 PM – Prob of arrival before 1 PM

$= (1 - e^{-2}) - (1 - e^{-1}) = (1 - 0.135) - (1 - 0.368) = 0.233$

c) What is probability that next arrival after 2 PM?

Reqd. Prob. = $P(X > 2) = e^{-2} = 0.135$

d) Nobody arrived till 1 PM. Probability of next arrival before 2 PM?

Using memory-less property, Reqd. Prob. = $1 - e^{-1} = 0.632$

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Queuing Example

In a parking lot on a Sunday, last arrival was at 12 noon. Arrival rate λ = 1/hour; service rate μ = 2/hour. No. of arrivals is Poisson and Inter-arrival time is exponential. Service time is also exponential.

e) Server is serving at 1 PM. Prob. that it is not complete by 2 PM?

μ = 2/hour. Using memory-less property, t = 1 hour
Reqd. Prob. = $e^{-2*1} = e^{-2} = 0.135$

f) Server is serving at 1 PM. Prob. that it is not complete by 1.10 PM?

μ = 2/hour. Using memory-less property, t = 10/60 = 1/6 hour
Reqd. Prob. = $e^{-2*1/6} = e^{-1/3} = 0.717$

f) Server is serving at 1 PM. Prob. that it will be complete by 1.10 PM?

μ = 2/hour. Using memory-less property, t = 10/60 = 1/6 hour
Reqd. Prob. = $(1 - e^{-2*1/6}) = (1 - e^{-1/3}) = 1 - 0.717 = 0.283$

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Queuing Example

Find probability of 2 or more arrivals in one hour when the number of arrivals are Poisson distributed with an arrival rate of 2 per hour.

Answer:

Poisson Distribution: Probability mass function $p(x) = (\lambda t)^x e^{-\lambda t})/x!$

Hence, Cdf of Poisson Distribution is given by: $F(x) = \sum_0^x (\lambda t)^n e^{-\lambda t})/n!$

Now customers are arriving at a rate of 2 per hour, i.e. $\lambda = 2$/hour.

From Poisson Distribution, $\lambda = 2$/hour; $t = 1$ hour, hence $\lambda t = 2$

Probability of 0 arrival, $p(0) = (2^0 e^{-2})/0! = e^{-2} = 0.135$

Probability of 1 arrival, $p(1) = (2^1 e^{-2})/1! = 2*e^{-2} = 0.271$

Hence, probability of 2 or more arrivals in next one hour

$= 1 - p(0) - p(1) = 1 - 0.135 - 0.271 = 0.594 = 59.4\%$.

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Birth and Death Process

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Birth-and-Death Processes

Birth and Death Process act as the foundation of commonly used queuing models

- **Birth – equivalent to arrival of a customer or job**
- **Death – equivalent to departure of a served customer or job**

Assumptions

1. **Given N(t) = n,**
   - **The time until the next birth (T_B) is exponentially distributed with parameter λ_n (Customers arrive according to a Poisson process)**
   - **The remaining service time (T_D) is exponentially distributed with parameter μ_n**

2. **T_B & T_D are mutually independent stochastic variables and state transitions occur through exactly one *Birth* ($n \rightarrow n+1$) or one *Death* ($n \rightarrow n-1$)**

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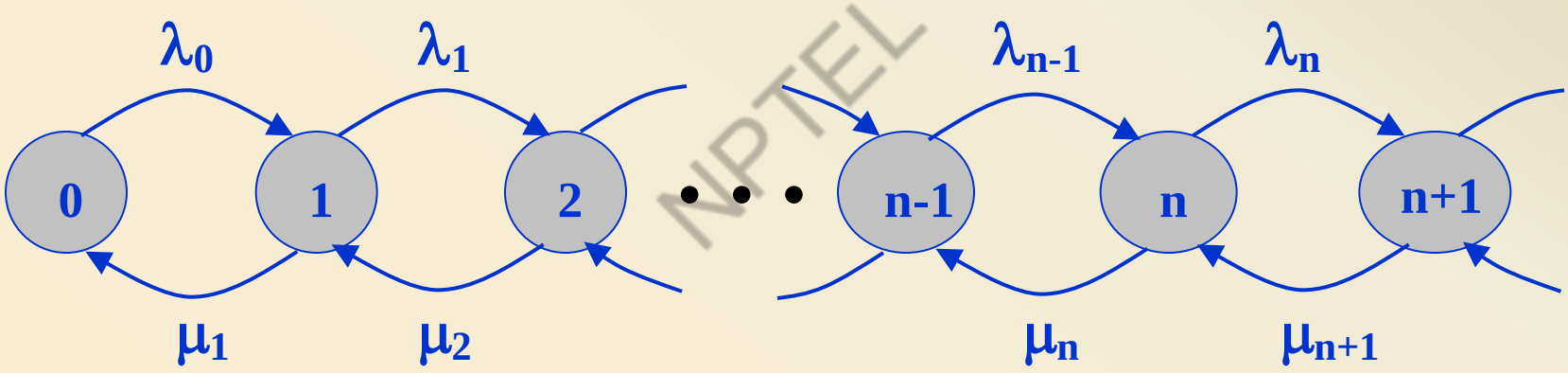
Birth-and-Death Process Rate Diagram

A Rate Diagram describes a Birth-and-Death process



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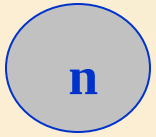

= **State n, i.e., the case of n customers in the system**

The M/M/1 Model

Assumptions - the Basic Queuing Process

- **Infinite input space and independence between arrivals**
- **The arrival process is Poisson with an expected arrival rate λ**
  - **Independent of no of customers currently in system**
- **A single queue is formed of no restriction on its length.**
- **No reneging or balking**
- **The queue discipline is FIFO**
- **The service mechanism consists of a single server with exponentially distributed service times**
  - **μ = expected service rate when the server is busy**

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Steady State Analysis

- **In steady state, the following balance equation must hold for every state n**

The Rate In = Rate Out Principle:

Mean entrance rate = Mean departure rate

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- **Also, the sum of all the probability states should be equal to 1.**

$$\sum_{i=0}^{\infty} P_i = 1$$

Steady State Analysis

State	Balance Equation		
0	$\mu_1 P_1 = \lambda_0 P_0$	$\Rightarrow$	$P_1 = \frac{\lambda_0}{\mu_1} P_0$
1	$\lambda_0 P_0 + \mu_2 P_2 = \lambda_1 P_1 + \mu_1 P_1$	$\Rightarrow$	$P_2 = \frac{\lambda_1}{\mu_2} P_1$
$\vdots$	$\vdots$		
n	$\lambda_{n-1} P_{n-1} + \mu_{n+1} P_{n+1} = (\lambda_n + \mu_n) P_n$	$\Rightarrow$	$P_n = \frac{\lambda_{n-1}}{\mu_n} P_{n-1}$
$\vdots$	$\vdots$		

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$$\text{Normalization}: \quad \sum_{i=0}^{\infty} P_i = P_0 \left(\underbrace{1}_{C_0} + \frac{\lambda_0}{\mu_1} + \underbrace{\frac{\lambda_0 \lambda_1}{\mu_1 \mu_2}}_{C_2} + \frac{\lambda_0 \lambda_1 \lambda_2}{\mu_1 \mu_2 \mu_3} + \cdots \right) = 1$$

Steady State Analysis

- **Steady State Probabilities**

$$P_0 = 1 \Big/ \sum_{i=0}^{\infty} C_i$$

$$P_n = C_n P_0$$

- **Expected No of Jobs in the System and in the Queue**
  - **Assuming c parallel servers**

$$L = \sum_{i=0}^{\infty} i \cdot P_i$$

$$L_q = \sum_{i=c}^{\infty} (i - c) \cdot P_i$$

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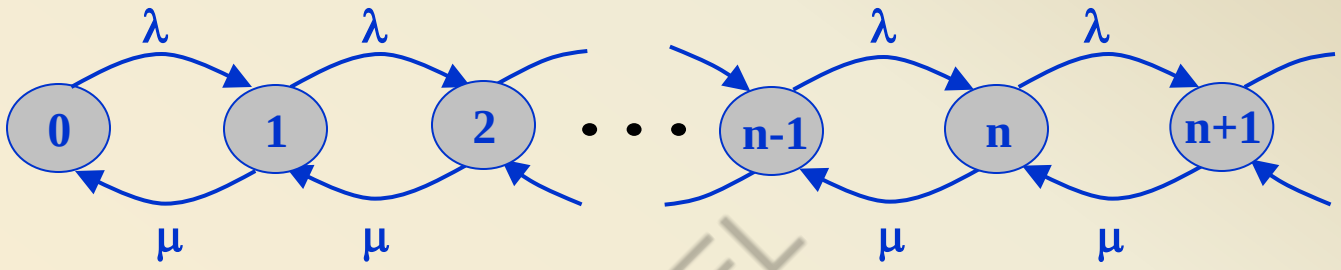
The M/M/1 Model

- $\lambda_n = \lambda$ and $\mu_n = \mu$ for all values of n=0, 1, 2, …



❖ **Steady State condition: $\rho = (\lambda/\mu) < 1$**

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$$P_0 = 1 - \rho$$

$$P_n = \rho^n (1 - \rho)$$

$$P(n \geq k) = \rho^k$$

$$L = \rho/(1-\rho)$$

$$W = L/\lambda = 1/(\mu - \lambda)$$

$$L_q = \rho^2/(1-\rho) = L - \rho$$

$$W_q = L_q/\lambda = \lambda/(\mu(\mu - \lambda))$$

Steady State Behavior of (M/M/1)

Deriving expression for probability of *n* customers in system, P_n

Avg arrival rate = λ

Avg service rate = μ

for stability, $\lambda < \mu$

The system state at any instant of time is the number of customers in the system.

Let $P_n(t)$: Probablity of n customers in the system at time t.

Assumption : Δt is very small during which there could be either 1 arrival or 1 departure or no arrival and no departure.

The probablity of system state at time $t + \Delta t$ can be found as

$$P_n(t+\Delta t) = P_n(t) \times \text{ probablity of no arrival during } \Delta t \times \text{probablity of no departure during } \Delta t$$

$$+P_{n-1}(t) \times \text{ probablity of one arrival during } \Delta t \times \text{probablity of no departure during } \Delta t$$

$$+P_{n+1}(t) \times \text{probablity of one departure during } \Delta t \times \text{ probablity of no arrival during } \Delta t$$

$$\Rightarrow P_n(t+\Delta t) = P_n(t)(1-\lambda\Delta t)(1-\mu\Delta t) + P_{n-1}(t)(\lambda\Delta t)(1-\mu\Delta t) + P_{n+1}(t)(\mu\Delta t)(1-\lambda\Delta t), \text{ for } n \geq 1$$

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Steady State Behavior of (M/M/1)

For $n \geq 1$

$$P_n(t+\Delta t) = P_n(t)(1-\lambda\Delta t)(1-\mu\Delta t) + P_{n-1}(t)(\lambda\Delta t)(1-\mu\Delta t) + P_{n+1}(t)(\mu\Delta t)(1-\lambda\Delta t)$$

$$\Rightarrow \frac{P_n(t+\Delta t) - P_n(t)}{\Delta t} = \lambda P_{n-1}(t) + \mu P_{n+1}(t) - (\lambda+\mu)P_n(t)$$

[Ignoring Δt^2 and higher order terms]

As $\Delta t \to 0$

$$\frac{dP_n(t)}{dt} = \lambda P_{n-1}(t) + \mu P_{n+1}(t) - (\lambda+\mu)P_n(t)$$

Under steady state:

$\frac{dP_n(t)}{dt} = 0$ and probability becomes independent of time, i.e. $P_n(t) \to P_n$

$$P_{n+1} = (1+\frac{\lambda}{\mu})P_n - \frac{\lambda}{\mu}P_{n-1}, \text{ for } n \geq 1$$

Let $\frac{\lambda}{\mu} = \rho < 1$ for stability

$$\boxed{P_{n+1} = (1+\rho)P_n - \rho P_{n-1}, \text{ for } n \geq 1 \qquad (1)}$$

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Steady State Behavior of (M/M/1)

For $n = 0$

(if system has no customer at time t there is no possibility of departure during next Δt)

$$P_0(t+\Delta t) = P_0(t) \times \text{Probablity of no arrival during } \Delta t$$
$$+P_1(t) \times \text{Probablity of one departure during } \Delta t$$

$$\Rightarrow P_0(t+\Delta t) = P_0(t)(1-\lambda\Delta t) + P_1(t)\mu\Delta t$$

$$\Rightarrow \frac{P_0(t+\Delta t) - P_0(t)}{\Delta t} = -\lambda P_0(t) + \mu P_1(t)$$

As $\Delta t \to 0$

$$\frac{dP_0(t)}{dt} = -\lambda P_0(t) + \mu P_1(t)$$

Under steady state, $\frac{dP_0(t)}{dt} = 0$ and $P_0(t) \to P_0, P_1(t) \to P_1$

$$\Rightarrow \lambda P_0 = \mu P_1$$

$$\Rightarrow P_1 = \frac{\lambda}{\mu} P_0$$

$$\boxed{P_1 = \rho P_0 \qquad (2)}$$

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Steady State Behavior of (M/M/1)

From (1) and (2), we have

$$P_{n+1} = (1+\rho)P_n - \rho P_{n-1}, \text{ for } n \geq 1$$

and

$$P_1 = \rho P_0$$

Thus,

$$n = 1, \quad P_2 = (1+\rho)P_1 - \rho P_0 = (1+\rho)\rho P_0 - \rho P_0 = \rho^2 P_0$$

$$n = 2, \quad P_3 = (1+\rho)P_2 - \rho P_1 = (1+\rho)\rho^2 P_0 - \rho^2 P_0 = \rho^3 P_0$$

$$\vdots$$

In general, $P_n = \rho^n P_0 \quad \text{for } n = 1, 2, \ldots$

$$\boxed{P_n = \rho^n P_0 \quad \text{for } n = 1, 2, \ldots}$$

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M/M/1 Queuing Model

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M/M/1-FCFS/Infinity/Infinity

Queuing system Classification example:

- **M/M/1/FCFS/infinity/infinity**
  - **M = Arrival process is Markovian**
  - **M = Service process is Markovian**
  - **1 = Number of servers.**
  - **FCFS = Queue Discipline First Come First Serve**
  - **Infinity = System size (infinite waiting line).**
  - **Infinity = Population size.**

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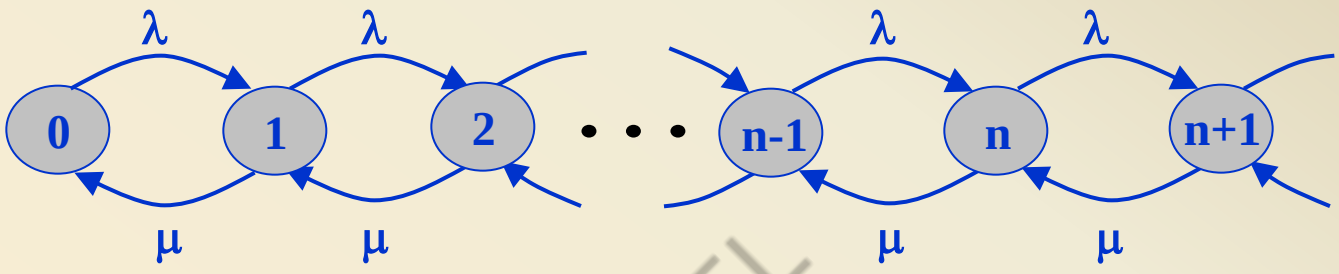
The M/M/1 Model

- $\lambda_n = \lambda$ and $\mu_n = \mu$ for all values of n=0, 1, 2, …



- **Steady State condition:** $\rho = (\lambda/\mu) < 1$

$P_0 = 1 - \rho$

$P_n = \rho^n(1-\rho)$

$P(n \geq k) = \rho k$

$L = \rho/(1 - \rho)$

$L_q = \rho^2/(1 - \rho) = L - \rho$

$W = L/\lambda = 1/(\mu - \lambda)$

$W_q = L_q/\lambda = \lambda/(\mu(\mu-\lambda))$

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Steady State Behavior of (M/M/1)

Deriving expression for probability of no customer in the system, P_0

Under steady state, system will be in either of the states $0, 1, 2, \ldots, n, \ldots, \infty$ with the corresponding probablities $P_0, P_1, P_2, \ldots, P_n, \ldots P_\infty$.

$$\therefore \sum_{n=0}^{\infty} P_n = 1$$

$$\Rightarrow P_0 + \sum_{n=1}^{\infty} \rho^n P_0 = 1$$

$$\Rightarrow P_0 (1 + \rho + \rho^2 + \ldots) = 1$$

$$\Rightarrow P_0 \frac{1}{1-\rho} = 1$$

$$\Rightarrow P_0 = 1 - \rho$$

$$\boxed{P_0 = 1 - \rho}$$

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Probability that the system is empty = Probability that the server is idle

=1− Probability that the server is busy

= P_0= 1 − ρ

ρ is called utilization factor or traffic intensity

Steady State Behavior of (M/M/1)

Average number of customers in the system, *L*

$$L = \sum_{n=0}^{\infty} nP_n$$

$$= \sum_{n=0}^{\infty} n\rho^n P_0 = P_0 \sum_{n=0}^{\infty} n\rho^n$$

$$L = P_0 \frac{\rho}{(1-\rho)^2} = (1-\rho)\frac{\rho}{(1-\rho)^2}$$

$$\Rightarrow L = \frac{\rho}{(1-\rho)} = \frac{\lambda}{\mu - \lambda}$$

To find the sum of series $\sum_{n=0}^{\infty} n\rho^n$

Let $S = \sum_{n=0}^{\infty} n\rho^n$, hence

$$S = \rho + 2\rho^2 + 3\rho^3 + \ldots\infty$$

$$\rho S = \qquad \rho^2 + 2\rho^3 + \ldots\infty$$

$$(1-\rho)S = \rho + \rho^2 + \rho^3 + \ldots\infty$$

$$\Rightarrow (1-\rho)S = \frac{\rho}{(1-\rho)}$$

$$\Rightarrow S = \frac{\rho}{(1-\rho)^2}$$

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Steady State Behavior of (M/M/1)

To find W, W_q and L_q

From Little's Law

Average time a customer spends in system:

$$W = \frac{L}{\lambda} = \left(\frac{\lambda}{\mu-\lambda}\right)\left(\frac{1}{\lambda}\right) = \frac{1}{\mu-\lambda}$$

Average waiting time:

$$W_q = W - \frac{1}{\mu} = \left(\frac{\lambda}{\mu-\lambda}\right) - \left(\frac{1}{\mu}\right) = \frac{\lambda}{\mu(\mu-\lambda)}$$

Average queue length:

$$L_q = \lambda W_q = \lambda \frac{\lambda}{\mu(\mu-\lambda)} = \frac{\lambda^2}{\mu(\mu-\lambda)}$$

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M/M/1 Example

Trucks arrive every 5 mins on the average. Average loading time is 3 mins. Assuming exponential inter-arrival and service times, find:

a) Prob. That a Truck has to wait, b) Total waiting time of a Truck, c) Expected Waiting time of Trucks in queue per day.

Answer: M/M/1 Queue and Exponential inter-arrival and service time,

Rate of arrival $\lambda = 60/5 = 12$ per hour

Rate of service $\mu = 60/3 = 20$ per hour

a) Utilization factor $\rho = \lambda/\mu = 12/20 = 0.6$.

Reqd Prob. = system is busy = 0.6

b) Total waiting time of a truck, $W = L/\lambda = (\rho/(1-\rho))*(1/\lambda)$

$= (0.6/(1-0.6))*(1/12) = 0.6/4.8 = 1/8$ Hour.

c) $W = 1/8$, So, $W_q = W - (1/\mu) = 1/8 - (1/20) = 3/40$ hour

No. of Trucks per day = 24 Hours per day * 12 per hour = 288

Exp Truck Waiting time in queue per day = (3/40)*288 = 21.6 Hour.

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M/M/1 Example

In a bank with a single server, there are two chairs for waiting customers. On an average, one customer arrives every 10 minutes and each customer takes 5 minutes for getting served. find:

- ***the probability that an arrival will find at least one chair free***
- ***the probability that an arrival will have to stand, and***
- ***expected waiting time of a customer in the queue.***

Answer:

Assumption: Poisson arrival and Exponential service time

This is a single server queuing system

Since, there are two chairs to wait, an arrival will find at least one chair free when

- **There is nobody in system**
- **There is one person in system and is in service (both chairs free)**
- **There are two persons in system, one of them is getting service (one chair is free)**

Hence, the required probability = p(0) + p(1) +p(2)

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M/M/1 Example

Probability that an arrival will find at least one chair free

- Average Interarrival time $1/\lambda = 10$ minutes,
- Average service time $1/\mu = 5$ minutes.
- Hence, the utilization factor, $\rho = \lambda/\mu = 5/10 = 0.5$

Now, $p(0) = 1 - \rho = 1 - 0.5 = 0.5$

- $p(1) = \rho * p(0) = 0.5*0.5 = 0.25$
- $p(2) = \rho * p(1) = 0.5*0.25 = 0.125$

So, the required probability = 0.5 + 0.25 + 0.125 = 0.875

Probability that an arrival will have to stand

An arrival will have to stand when not even one chair will be found.

So, the required probability in this case will be given by,

= 1 – probability that the arrival finds at least one chair free

= 1 – 0.875

= 0.125

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M/M/1 Example

<u>Expected waiting time of a customer in the queue (Wq)</u>

Expected Length of the System $L = \rho/(1 - \rho)$

Expected Queue length $Lq = L - \rho$

Expected Waiting Time of a Customer in the queue $Wq = Lq/\lambda$

Average Interarrival Time = $1/\lambda$ = 10 minutes

Utilization Factor $\rho = 0.5$

Hence, we have

$L = 0.5/(1 - 0.5) = 1;$

$Lq = 1 - 0.5 = 0.5;$

$Wq = 0.5*10 = 5$ minutes.

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Another M/M/1 Example

Arrivals to an airport with a single runway are poisson distributed with a rate of 30 per hour. The average time to land an aircraft is 90 seconds and this time is exponentially distributed. Find the utilization of the runway and the steady state parameters L (Length of system) and W (Average waiting time in the system).

Answer:

Arrival rate λ = 0.5 per minute (Arrival is 30 per minute)

Service rate μ = 60/90 per minute = 2/3 per minute.

Runway utilization (with a single server M/M/1 queue)

$\rho = \lambda / \mu = (1/2)/(2/3) = 3/4 = 75\ \%$

Steady State Parameters:

Length of system $L = \rho / (1 - \rho) = (3/4)/(1 - 3/4) = (3/4)/(1/4) = 3$ aircrafts.

Average waiting time in system: $W = L / \lambda = 3/(1/2) = 6$ minutes.

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Queuing Problems

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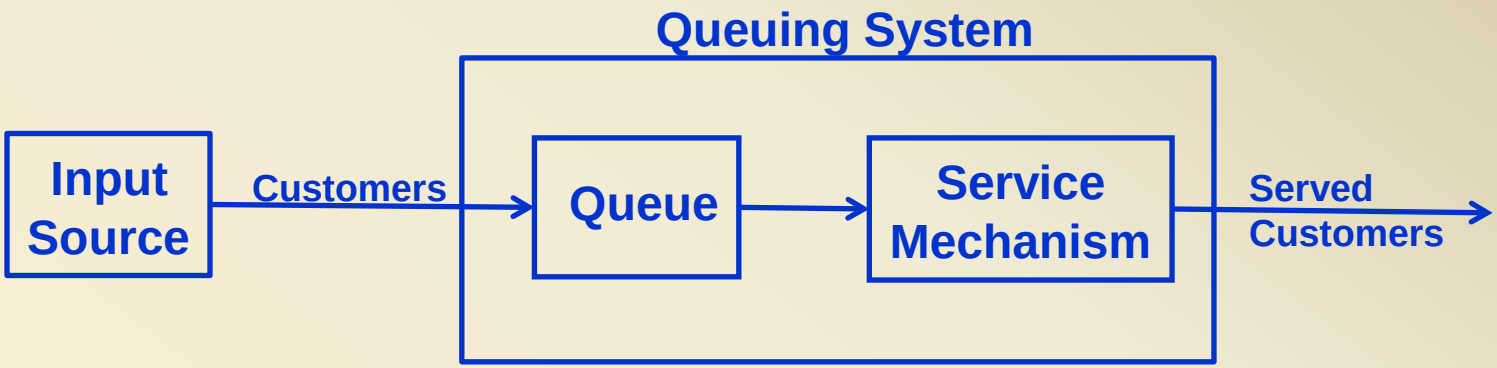
A Note on M/M/1 Arrival and Service



Arrival

- The **mean rate of arrival** of entities at a service station is λ per hour
- But **actual number of arrivals** in any hour follows a **Poisson distribution**.

Service

- The **mean service time** of the service station is (1/μ) hour.
- But the **actual service time** during the service period follows an **exponential distribution**.
- In another language, the **actual number** being serviced per hour follows **Poisson distribution** and the mean service rate is μ /hour.

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Comparison of L, Lq and Lq'

For M/M/1 Queuing System:

L: Average number of customers in the system $$L = \sum_{n=0}^{\infty} n \cdot P_n = \frac{\rho}{1-\rho}$$

Lq: Average number of customers in the queue

$$L_q = \sum_{n=1}^{\infty} (n-1) \cdot P_n = L - \rho = \frac{\rho^2}{1-\rho}$$

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Lq': Average number of customers in non-empty queue

$$L_q^{'} = \sum_{n=2}^{\infty} (n-2) \cdot P_n^{'} = \frac{L_q}{1-P_0-P_1}$$ **because** $$P_n^{'} = \frac{P_n}{1-P_0-P_1}$$

Hence,

$$L_q^{'} = \frac{L_q}{1-(1-\rho)-\rho(1-\rho)} = \frac{L_q}{1-1+\rho-\rho+\rho^2} = \frac{\rho^2}{(1-\rho)\rho^2} = \frac{1}{(1-\rho)}$$

Comparison of L, Lq and Lq’

In a service centre, customer arrive randomly at 10 per hour following Poisson distribution. There is only one server. The service time is exponential and the mean service time is 5 minutes per customer. Find:

a) Average Number of customers in the system (L).
b) Average Number of customers in the queue (Lq).
c) Average number of customers in non-empty queue (Lq’)

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Comparison of L, Lq and Lq'

In a service centre, customer arrive randomly at 10 per hour following Poisson distribution. There is only one server. The service time is exponential and the mean service time is 5 minutes per customer. Find:

a) Average Number of customers in the system (L).
b) Average Number of customers in the queue (Lq).
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Arrival Rate per hour (λ)	Service Rate per hour (μ)	Utilization Factor ($\rho = \lambda/\mu$)	Average No. of customers in the system $L = \rho/(1 - \rho)$	Average No. of customers in the queue $Lq = \rho^2/(1 - \rho)$	Average No. of customers in non-empty queue $Lq' = 1/(1 - \rho)$
10	60/5 = 12	10/12 =5/6	(5/6)/(1–5/6) = 5	$(5/6)^2/(1–5/6)$ = 25/6 = 4.166	1/(1-5/6) = 6

Why is Lq' greater than Lq?

An M/M/1 Problem

- ***Arrival of planes is Poisson at an airport with a mean rate of 20 per hour. The airport can land 60 planes per hour, on an average, in good weather and 30 planes per hour in bad weather with a Poisson service rate. When there is congestion, the planes are forced to fly over the field in the stack awaiting landing.***
- ***How many planes should be flying over the field in the stack on an average in good weather and in bad weather?***
- ***How long a plane would be in the stack on the average in good weather and in bad weather?***

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An M/M/1 Problem

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- ***How many planes should be flying over the field in the stack on an average in good weather and in bad weather?***
- ***How long a plane would be in the stack on the average in good weather and in bad weather?***

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Weather	Arrival Rate per hour (λ)	Service Rate per hour (μ)	Utilization Factor ($\rho = \lambda/\mu$)	Average No. of Planes in Stack $L_q = \rho^2/(1-\rho)$	Average Waiting Time is Stack $W_q = L_q/\lambda$
Good	20	60	1/3	$(1/3)^2/(1-1/3) = (1/9)*(3/2) = 1/6$	$(1/6)/20 = 1/120 = 1/2$ min
Bad	20	30	2/3	$(2/3)^2/(1-2/3) = (4/9)*(3) = 4/3$	$(4/3)/20 = 1/15 = 4$ min

Another M/M/1 Problem

In a service centre only one person can get service. Any other arriving customer has to wait outside as there is no place available inside. The customers arrive randomly at 8 per hour. The service is exponential and takes 6 minutes on the average. Find the probability that an arriving customer will have to wait outside.

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Another M/M/1 Example

In a service centre only one person can get service. Any other arriving customer has to wait outside as there is no place available inside. The customers arrive randomly at 8 per hour. The service is exponential and takes 6 minutes on the average. Find the probability that an arriving customer will have to wait outside.

Answer:

Average arrival rate $\lambda = 8$ per hour.

Average service time $1/\mu = 6$ minutes, and hence $\mu = 10$/hour.

Hence, the utilization factor, $\rho = \lambda/\mu = 8/10 = 4/5$

So, the probability that an arrival will find the place free

$= p(0) = 1 - \rho = 1 - (4/5) = 1/5$

An arriving customer has to wait outside when there is no place free inside. So, probability that an arriving customer has to wait outside:

= 1 – probability that the arrival finds the place free

$= 1 - (1/5) = 4/5 = 80\%$.

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THANK YOU

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